2021-Summer Paper 4 – Recent Advances

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Q1. Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 2, Q3f)

"Focal" is taken to mean localised (organ-confined and locally advanced) disease.

Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**.

EBRT (external beam radiotherapy) techniques

Step	Advance
3D-CRT (three-dimensional conformal RT)	CT-planned beams shaped to the prostate
IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)	Inverse planning, steep dose gradients; less rectal toxicity; now standard
IGRT (image-guided RT)	Daily fiducials or cone-beam CT (computed tomography); margins 3–7 mm
MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography)	Better targeting; PSMA PET changes plans in up to half of high-risk men

Schedule	Dose per fraction	Example	Evidence
Conventional	1.8–2 Gy	78 Gy in 39	Dose escalation (RTOG 0126): better PSA control, no survival gain
Moderate hypofractionation	2.25–4 Gy	**60 Gy in 20**	**CHHiP, PROFIT**: non-inferior
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity

Brachytherapy

Feature	LDR (low dose rate)	HDR (high dose rate)
Source	Permanent ^{125}I (145 Gy) or ^{103}Pd seeds	Temporary ^{192}Ir via afterloader

Feature	LDR (low dose rate)	HDR (high dose rate)
Monotherapy	Low and favourable intermediate risk	27 Gy in 2 or 34–38 Gy in 4 (single 19 Gy fraction fails)
Quality	Post-implant CT: **D90** (dose covering 90% of gland) **>90%** of prescription	Real-time planning

- **Technique:** TRUS (transrectal ultrasound)-guided perineal template; peripheral loading spares the urethra.
- **Boost** (EBRT 45 Gy + implant) for unfavourable intermediate/high risk: **ASCENDE-RT** halved biochemical failure, with more grade 3 urinary toxicity (18% vs 5%). **RTOG 0232:** adding EBRT to implant gave no benefit in intermediate risk.
- **Contraindications:** IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s, recent TURP (transurethral resection of the prostate), pubic arch interference, prior pelvic RT, inflammatory bowel disease.

ADT (androgen deprivation therapy) with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

Risk group	ADT with EBRT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High/locally advanced	**18–36 months** (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE)

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

Advanced and metastatic disease

Setting	RT role	Evidence
Low-volume de novo mHSPC (metastatic hormone-sensitive)	**RT to prostate** + ADT	**STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume
Oligometastases (≤3–5)	SBRT to all lesions (MDT, metastasis-directed therapy)	STOMP, **ORIOLE** (less progression), SABR-COMET
Painful bone metastases	Palliative EBRT, e.g. **8 Gy single fraction**; cord compression	Equal pain relief to multiple fractions
mCRPC (metastatic castration-resistant), bone-only symptomatic	**Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6	**ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events

Setting	RT role	Evidence
mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor)	**^{177}Lu-PSMA-617** (β-emitter radioligand)	**VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL. **Benign bounce** after brachytherapy in about a third: watch, do not salvage.
- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

Toxicity

Type	Features and management
Acute (weeks)	Frequency, dysuria, diarrhoea; α-blockers
Late GI (gastrointestinal)	**Radiation proctitis** at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen
Late GU (genitourinary)	Cystitis, haematuria, stricture
Sexual	ED (erectile dysfunction), vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most
Rare	Second cancers, fistula

Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; PSMA radioligands moving earlier (PSMAfore).

Diagram to draw: TRUS-guided perineal template brachytherapy.

Key points to remember:

- α/β ~1.5 Gy: basis of 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.

Q2. Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival.

Type: LAQ (25 marks) (also answered in 2019-Summer Paper 1, Q3d and 2019-Summer Paper 2, Q3c)

Definition

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).

Classification

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary disease, **exogenous testosterone**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone)
Testicular: **NOA (non-obstructive azoospermia)**	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps	Small testes, high FSH
Post-testicular: **OA (obstructive azoospermia)**	CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation (brief)

- Absent vasa → CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL → post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm → obstruction likely;** FSH >7.6 and axis ≤4.6 cm → spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

Testicular biopsy

Indications

- Azoospermia with **normal testes, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI (intracytoplasmic sperm injection) needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present).

Types

Type	Method	Use	Limits
Open biopsy	Small incision in the albuginea, pea-sized tissue	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core biopsy	14 G biopsy gun under cord block	Fresh retrieval in OA	Blind; may injure epididymis or testicular artery; avoid in scarred scrotum

Type	Method	Use	Limits
FNA (fine-needle aspiration), i.e. TESA (testicular sperm aspiration)	23 G needle, suction, several passes	OA retrieval; backup to epididymal methods	Few tubules, poor architecture; low yield in NOA
Microdissection biopsy (micro-TESE)	Operating microscope	NOA	Long operation

- **Also know:** multi-site FNA "mapping" in NOA.

Open biopsy technique

1. Local anaesthesia with cord block (or general/spinal); choose the larger testis.
2. 2 cm transverse incision; open tunica vaginalis.
3. 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch or wet prep** at once for sperm; negative → contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

Histology patterns

Pattern	Meaning
Normal	Suggests obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis/hyalinisation	End-stage (e.g. Klinefelter)

- **Complications:** haematoma (commonest); **inadvertent epididymal biopsy** (most serious: causes obstruction); infection rare.

Surgical sperm retrieval

Indications: CBAVD; unreconstructable obstruction; declined or failed vasectomy reversal; backup at reconstruction; NOA; anejaculation.

Technique	Method	Best for	Notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput	OA	Few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Many motile sperm, multiple frozen vials
TESA	23 G needle, suction	OA	Least invasive; immotile sperm
Conventional TESE	Open biopsy, one or more sites	OA, NOA	Blind sampling
Micro-TESE (Schlegel 1999)	Testis opened in an avascular plane; **larger, opaque tubules** excised under the microscope	**NOA (best)**	Most sperm, least tissue

Key principles

- **MESA:** start distally and move **proximally** until motile sperm appear; the most motile sperm are often in the efferent ductules.

- **Micro-TESE:** spermatogenesis in NOA is **focal**. Thin white tubules are Sertoli cell–only; plump ones hold sperm. Embryologist examines tissue in theatre; map positive sites; negative → other testis.

Micro-TESE retrieval by diagnosis

Condition	Retrieval
Post-orchidopexy	74%
AZFc deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
SCO	30%
AZFa/AZFb deletion	**0%**

- Overall sperm found in **about half** of NOA.
- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

Use of retrieved sperm

- **ICSI is essential** (poor motility). Frozen sperm equal fresh, except in spinal cord injury.
- NOA: more miscarriage (sperm DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- **Also know:** post-mortem retrieval needs prior consent; microfluidic sperm sorting.

Diagram to draw: micro-TESE: bivalved testis with plump spermatogenic tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
> - Azoospermia: no sperm in centrifuged pellet on two samples.
> - Open biopsy is the gold standard; fix in Bouin's, never formalin.
> - OA: MESA; NOA: micro-TESE (sperm in about half).
> - AZFa/b deletion: 0% retrieval, do not operate.

Q3a. Operating room safety : the role of surgeon.

Type: SAQ (10 marks)

Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

Phase	Key checks
Sign in (before induction)	Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL
Time out (before incision)	Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed
Sign out (before leaving theatre)	Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; FiO_2 (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.
8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

Diagram to draw: Swiss cheese model with aligned holes.

> **Key points to remember:**
> - WHO checklist: sign in, time out, sign out.
> - Universal Protocol: verify, mark, time-out.
> - Count discrepancy: ~100-fold risk of a retained item.
> - Fire triangle: surgeon owns ignition.

Q3b. Detrussor underactivity.

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 2, Q3d)

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** a contraction of reduced strength and/or duration, causing prolonged or incomplete emptying. **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex (slow stream, hesitancy, straining, incomplete emptying). DUA is common in elderly men and women having urodynamics.

Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

Type	Examples
Neurogenic	Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, radical pelvic surgery
Myogenic	Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing
Drugs	Anticholinergics, opioids
Functional	Fowler syndrome, dysfunctional voiding

Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if deficit, drug review.
- **Pressure-flow study is diagnostic:** low detrusor pressure with low flow.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen.

Management

- **Goals:** protect kidneys; prevent UTI (urinary tract infection) and stones; relieve symptoms. No drug reliably improves contraction.

Aim	Options
Drainage	Timed and double voiding; **CISC (clean intermittent self-catheterisation), the mainstay**; suprapubic catheter if CISC impossible
Lower outlet resistance	α-blockers; pelvic-floor biofeedback; outlet surgery (TURP, transurethral resection of the prostate) in selected obstructed men
Improve contraction	**SNM (sacral neuromodulation)** for non-obstructive retention (Fowler); **sacral anterior root stimulator** with rhizotomy in complete spinal injury
Not recommended	Bethanechol, distigmine (weak evidence; bronchospasm, rarely cardiac arrest); Credé or Valsalva voiding (reflux risk)
Experimental	Intravesical prostaglandins, intravesical electrotherapy, TAC-302, latissimus dorsi myoplasty, stem cells

- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

Diagram to draw: pressure-flow plot with BCI lines at 100 and 150.

Key points to remember:

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; parasympathomimetics not recommended.
- SNM for Fowler syndrome.

Q3c. Surgical management of primary scrotal cancer.

Type: SAQ (10 marks)

Introduction

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes**, crossing the midline: both groins at risk.

Evaluation

- Nodule or ulcer on the anterolateral scrotum, often late; **biopsy** confirms.
- Examine both groins; FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) for staging.

Staging (Lowe modification of Ray and Whitmore)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes only
C	Pelvic nodes, no distant spread
D	Distant metastases

Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded. Hemi- or total scrotectomy for large tumours.
3. **Reconstruction:** scrotal laxity usually allows primary closure. Defect >50%: local or thigh flaps, split-skin graft, or thigh pouches for the testes.
4. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.

Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours. Routine prophylactic dissection is not advised.
- **Bulky/fixed nodes:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.

Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC (excision or Mohs); melanoma (excision by Breslow depth ± sentinel node); sarcoma (wide excision).

Diagram to draw: elliptical wide excision with 2 cm margin and inguinal node drainage.

Key points to remember:

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

Q3d. Painful bladder syndrome management.

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 2, Q3b)

Definition

- **IC/BPS (interstitial cystitis/bladder pain syndrome), AUA (American Urological Association):** unpleasant bladder-related sensation (pain, pressure, discomfort) with LUTS (lower urinary tract symptoms) for **>6 weeks**, without infection or other cause. ESSIC (European Society for the Study of Interstitial Cystitis) uses >6 months.
- **Phenotypes:** Hunner lesion (HL) disease (true inflammation) versus non-HL; also pelvic-floor-dominant and widespread (centralised) pain.

Before treatment

- Exclude UTI (urinary tract infection), CIS (carcinoma in situ), stones, **ketamine cystitis**, OAB (overactive bladder), radiation cystitis; PVR (post-void residual), urinalysis, culture.
- **Office cystoscopy** if over 50 or HL suspected.

Principles (AUA 2022)

- **Old AUA (2011/2014): six tiers**, from self-care to surgery. **AUA 2022 dropped the tiers:** phenotype-guided, shared decisions, multimodal care; major surgery stays last.

Treatment options

Group	Options
Behavioural	Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plans
Physical	**Myofascial pelvic-floor physiotherapy** if tender (Standard); **avoid Kegel exercises**
Oral	**Amitriptyline** 10 mg at night, up to ~50 mg; hydroxyzine; cimetidine; **PPS (pentosan polysulfate)** 100 mg three times daily (**pigmentary maculopathy**: retinal checks)
Intravesical	**DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min**; heparin; alkalinised lidocaine (flares)
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 cmH_2O) hydrodistension; **onabotulinumtoxinA 100 U** (risk of retention); SNM (sacral neuromodulation)
Refractory HL	**Ciclosporin A**
Last resort (<10%)	Diversion ± cystectomy, or augmentation; pain may persist

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Monitor with symptom scores; stop ineffective treatments. Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS (irritable bowel syndrome), fibromyalgia) with other specialists.

Diagram to draw: cystoscopic Hunner lesion with radiating vessels.

> **Key points to remember:**
>
> - AUA 2022: no tiers; phenotype-based.
> - HL: fulguration/triamcinolone; ciclosporin if refractory.
> - PPS: maculopathy; antibiotics and BCG not offered.

Q3e. Secondary tumors after urinary diversion.

Type: SAQ (10 marks)

Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more). Distinct from recurrent urothelial cancer.

Risk by diversion (German registry, Kälble 2011)

Diversion	Risk	Surveillance
Ureterosigmoidostomy	2.6% (6–29% in long-term series)	Regular colonoscopy **from 5 years**
Augmentation cystoplasty	1.6%	Endoscopy, often from 10 years
Colonic neobladder	1.3%	—
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder	0.05%	No routine
Ileal conduit	0.02%	No routine

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis.
- **N-nitrosamines** from bacterial action on urinary nitrates (probably enhancers, not sole cause); chronic inflammation; growth factors (EGF, epidermal growth factor); ornithine decarboxylase; abnormal mucin.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

Pathology

- Mostly **adenocarcinoma**; also adenomatous polyps, urothelial carcinoma, sarcoma.
- **Augmentation:** adenocarcinoma and aggressive urothelial carcinoma, often advanced at diagnosis; earliest at 4 years. **Gastrocystoplasty** carries relatively high risk. Exstrophy and immunosuppression add risk.
- **Nephrogenic adenoma:** benign; commonest tumour after cystoplasty.

Presentation

- Haematuria, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent infection, mass.

Management

- Biopsy and staging.
- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total cystectomy with new diversion.
- **Prevention:** prefer ileal segments; counsel patients; lifelong follow-up.

Also know: after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance.

Diagram to draw: ureterosigmoid anastomosis with tumour at the urothelial–colonic junction.

> **Key points to remember:**
>
> - Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
> - Mostly adenocarcinoma at the anastomosis.
> - Excise defunctioned anastomoses.

Q3f. Role of open and laparoscopic stone surgeries in modern era of endourology.

Type: SAQ (10 marks)

Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

Indications

Category	Examples
Endourology failed or unlikely	Very complex staghorn needing many PCNL sessions; failed multiple procedures
Anatomical problem treated together	**PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis
Difficult access	Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity
Poor renal function	**Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole
Large impacted ureteric stone	Laparoscopic ureterolithotomy after failed URS/SWL
Concomitant surgery	Other open or laparoscopic procedure
Bladder	Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture

Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

Feature	Open	Laparoscopic/robotic
Morbidity	Large incision, pain, long stay	Less pain, shorter stay
Kidney	Ischaemic loss of function (anatrophic)	Similar repair, less wound morbidity
Later surgery	Adhesions make redo difficult	Fewer

Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

Diagram to draw: Brödel's line incision for anatrophic nephrolithotomy.

Key points to remember:

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

Rapid Revision Sheet – 2021-Summer Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate? (LAQ)

- α/β ~1.5 Gy: basis of 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.

Q2. Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival. (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Open biopsy is the gold standard; fix in Bouin's, never formalin.
- OA: MESA; NOA: micro-TESE (sperm in about half).
- AZFa/b deletion: 0% retrieval, do not operate.

Q3a. Operating room safety : the role of surgeon. (SAQ)

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Q3b. Detrussor underactivity. (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; parasympathomimetics not recommended.
- SNM for Fowler syndrome.

Q3c. Surgical management of primary scrotal cancer. (SAQ)

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

Q3d. Painful bladder syndrome management. (SAQ)

- AUA 2022: no tiers; phenotype-based.
- HL: fulguration/triamcinolone; ciclosporin if refractory.
- PPS: maculopathy; antibiotics and BCG not offered.

Q3e. Secondary tumors after urinary diversion. (SAQ)

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis.
- Excise defunctioned anastomoses.

Q3f. Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.